SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: [Insert date]

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick “No” or write “None” – do not leave it blank. Submit as a single PDF document named “SDC2026_TeamName_TechDeclaration.pdf” together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

pastaisfine

Project / Solution Name

Roast my CV

Name (Leader)

Darren Tan Yik Wei

Email & Phone (Leader)

darrentyw5955@gmail.com

University Affiliation(s)

Universiti Teknologi Malaysia

Live Prototype URL

https://roast-my-cv-seven.vercel.app/

Code Repository URL (public / access granted to judges)

https://github.com/pastaisfine/roast-my-cv-sdc-hack.git

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

The React web app handles CV upload, tone selection (gentle, standard, blunt), the Future Skills Radar treemap and SDG views. It calls an Express.js REST API running on Node.js, deployed on Replit. The API extracts text from the uploaded CV and sends it to a RunPod serverless endpoint, which returns embeddings. Role matching computes cosine similarity between the CV embedding and occupation skill-profile embeddings. Occupation data comes from public job-posting and labour-market data from DOSM, held in PostgreSQL. A scheduled job re-scores occupations after each DOSM snapshot and regenerates a compiled knowledge-context file. The API then sends the CV text, the matched roles and that context file to DeepSeek V4.1 Flash, accessed through OpenCode. The model generates the roast, the AI exposure rationale, moderate versus fast disruption scenarios and the pathway text. Each recommendation shows its data source and date. Users see predictions as estimates, and the app supports career decisions without making hiring decisions. [Attach block diagram as appendix.]

2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
JavaScript (ES2022) / Node.js [version]	Frontend and backend runtime	MIT
React [version]	Frontend UI	MIT
Express.js [version]	Backend REST API	MIT
multer, pdf-parse	CV file intake and text extraction	MIT
D3 or Recharts	Future Skills Radar	MIT License• BSD 3-Clause License

3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Gemini 3.6 Flash	Roast, exposure rationale, scenarios, pathway and SDG draft text	Free
RunPod Serverless endpoint	Embedding generation for CV-to-role matching	Sponsor-provided (US$25 team credit)
DOSM open data API / downloads (open.dosm.gov.my)	Labour-market demand data refreshed per snapshot	Free

3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Development of frontend and partial backend	Replit Core (sponsor-provided)
RunPod	GPU endpoint for embeddings	Serverless
GitHub	Source control and public repository	Free

4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Gemini 3.6 Flash	Google AI Studio	Pre-trained, no fine-tuning	Does not require license
Embedding model intfloat/multilingual-e5-small	run on RunPod	Pre-trained	MIT
EasyOCR Model	run on RunPod	Pre-trained	Apache License Version 2.0

4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
DOSM labour-market and job data	open.dosm.gov.my	Public	CC BY 4.0	N
MASCO occupation classification (groups 1 to 5)	https://emasco.mohr.gov.my/	Public	CC BY 4.0	N
Karpathy jobs repo (reference for exposure score design)	https://github.com/karpathy/jobs	Public	No License	N
UN SDG targets and indicators	https://sdgs.un.org/	Public	Free/No License	N
Sample CV profiles for demo	Self-authored	Synthetic	Own work	N

4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Replit AI	Web App Development, code completion	Major
Claude ai	Brainstorming, PRD drafting, declaration drafting, code and debugging help	Major

4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The idea, problem framing and design decisions came from me. We reviewed Andrej Karparthy's Job Visualisation repo and edited all AI output before use rather than adopting it wholesale. We ran and tested generated code ourselves, and manually checked every row of the occupation-to-SDG table and every course in the pathway list against its provider's site. The app does not generate course names or statistics freely. Pathway courses come from a verified short list, and demand figures come from DOSM data. I can explain every part of the codebase.

5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit AI	Replit Core subscription	Web app Development
RunPod	US$25 credit	GPU endpoint for embedding generation, and OCR Model

5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit is compulsory for hosting but replaceable, since the Express and React app runs on any Node.js host. RunPod is used for embeddings and hosting OCR Model. These could be replaced by a hosted embedding API or a CPU-run small model, with slower processing and some loss of multilingual quality. Neither sponsor resource contains application logic or data, so core functionality does not depend on them.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The app processes CV text and the user's chosen tone, and sends the text to the Gemini 3.6 Flash by Google AI Studio to the RunPod embedding endpoint. Demo use relies on synthetic CVs only, with no real personal data. [CV text is held in memory for the request and is not written to the database; no names, emails or phone numbers are stored.] Transport uses HTTPS. API keys are held in Replit Secrets and are not committed to the repository. [no CV data retained after the session.] Users are told that their CV text is sent to third-party AI services before they submit it.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The language model may hallucinate jobs, courses or statistics. Mitigation: occupations come from the MASCO and DOSM tables, courses from a verified short list, and every recommendation cites a source and date. The exposure scores and scenarios are LLM-derived estimates, and the interface labels them as such. Embedding similarity may carry language or background bias, so the skill-fit score is shown with its skill gap breakdown, and the app does not make hiring decisions. The prototype depends on a single LLM provider and a single embedding endpoint; the fallback is a swap to an alternative hosted model. Scope covers MASCO groups 1 to 5 only. Rate limits on the free tiers may slow the demo, so common results are cached.

7. COMPLIANCE & READINESS

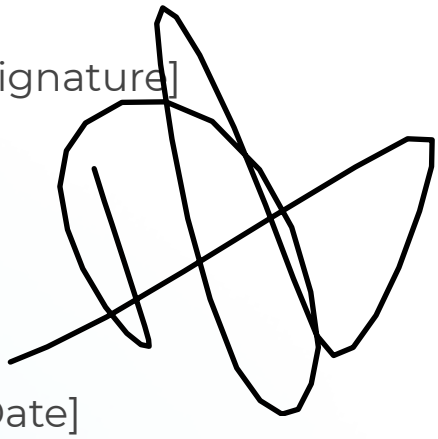
Tick all boxes that apply before submitting.

- [x] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- [x] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- [x] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- [x] Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- [x] Any use of official SDC sponsor resources has been declared in Section 5.
- [x] The prototype described here is accessible to judges through the links provided in Section 1.
- [x] All fields in this template have been completed within the word limits stated.

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]



[Date]

23 Oct 2026